VORONOI-BASED INDOOR ROOM SEARCH AND RESCUE

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This technical report is to process of my research during the Summer, 2021.
Technical Report

***Abstract*—In this project, we investigate of autonomous robots for mapping, navigation, with the shortest path planning to the target. We implemented Voronoi Diagram for path planning such as the Global Path and avoiding the obstacles, we generated local points (multiway-points) and applied Travel Salesman Problem (TSP) Algorithms to find the optimal path (shortest path planning) to connect the waypoints by searching the closest edge of the voronoi diagram, we then applied the Dynamic Window Approach method (DWA) to allow the robot to navigate locally based on the TSP and voronoi diagram paths. The robot was given an assignment to navigate from the start point (source) to the end point with local points (targets). Then the robot has to complete the assignment as the followings from the start point find the shortest path, go to the local points (the targets), with avoiding the obstacles.**

Index Terms:

I. PROJECT OUTLINE

SEARCH and rescue is the final goal of this research project. Assume we have a disaster, and we want to search people whom were injured inside a building and we want the robot to navigate and search inside the rooms. To do this we need to find the path planning, determine the targets, find the optimal path for the target with shortest distant, then navigate locally to the targets. This can be categorized as four stages as the followings: Environmental Exploration, Multi-Goal Path Planning, Local Navigator, and 2D Grid Mapping.

In the Environmental Exploration stage, we constructed two kinds of Voronoi Diagram maps: Voronoi Diagram, and Generalized Voronoi Diagram (GVD) as follows as it shows in fig. 1. and fig. 2.

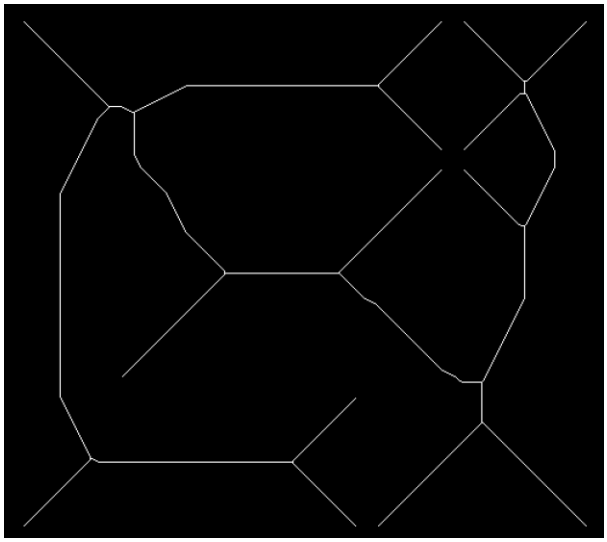


Fig. 1. Generalized Voronoi Diagram (GVD

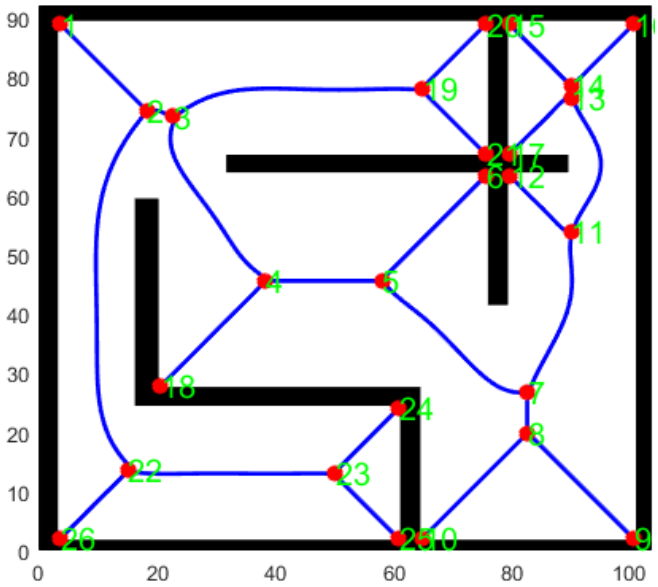


Fig. 2. Voronoi Diagram with the path planning

In the Multi-goal path planning, we used the Voronoi Diagram (GVD) maps with for TSP (Travelling salesman problem) algorithms to connect some goals/waypoints that we need to explore, rescue,

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and reach. Running your TSP algorithm to connect some points such as the green circles P_n in Fig. 3. TSP can connect waypoints, $P_0, P_1, P_2 \ldots P_n$ (injured persons – we have known their GPS coordinates such as the local points in Table. 1) in certain sequence to minimize the total traveling distance. The result would be the optimal path, the green path in Fig. 3, for the targets (local points).

TABLE I
THE LOCAL POINTS FOR TSP

Local locations	Local Points	X	Y
0	P_0	35	33
1	P_1	258	73
2	P_2	257	230
3	P_3	428	307

TSP local points as we chose for the targets

The red points are the Voronoi diagram points where number 1 is the start points and number 25 is the end point.

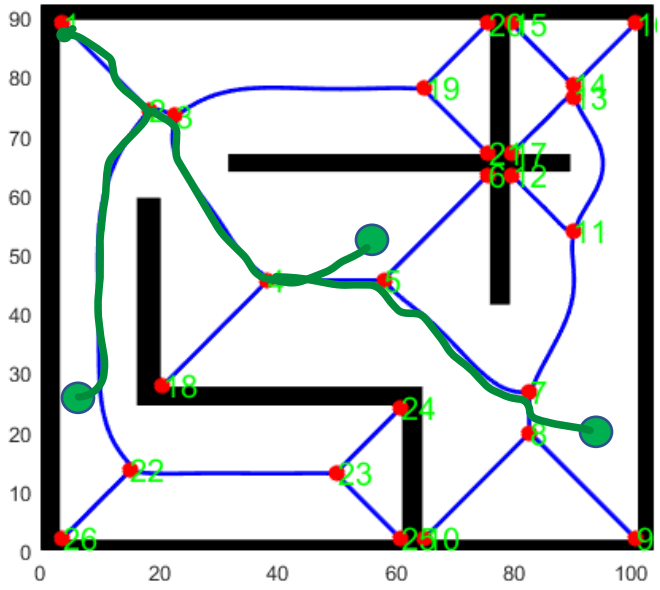


Fig. 3 the expected optimal path using TSP (Travelling salesman problem) algorithms. the blue line is the path planning for the Voronoi Diagram, the green path is the optimal path for the targets (local points), and the green circles are the targets.

Once sequence has been determined, robot needs to carry out path planning (navigation).
• Robot will perform navigation/path planning.
• Build up map (Mapping).

The local navigator we implemented in this project is the DWA (Dynamic Window Approach) method to walk a mobile robot from starting point, the red point number 1, to goal, the P_n green points, along the edges of built GVD, the blue path. This task significantly to using DWA to navigate the robot to connect multiple goals depicted by the green points such that the green path in Fig. 3.

II. THE PROJECT STAGES

1) Environmental Exploration

We constructed the Voronoi Diagram using the map in Fig. 4. The map black lines are the walls (obstacles). The robot can only move in the empty area in the map and avoid any obstacles such as the walls. To find the global path, we used the Voronoi Diagram method as it shows in Fig. 2. In this stage we must specified the robot radius, the obstacles, the board. The radius of the robot, is used to compute configuration free space as it shows in Fig. 5. in case if the robot's radius is big, the path in the graph will be disconnected which means it can not pass through a narrow corridor or door in the map due to the collision's expectations.

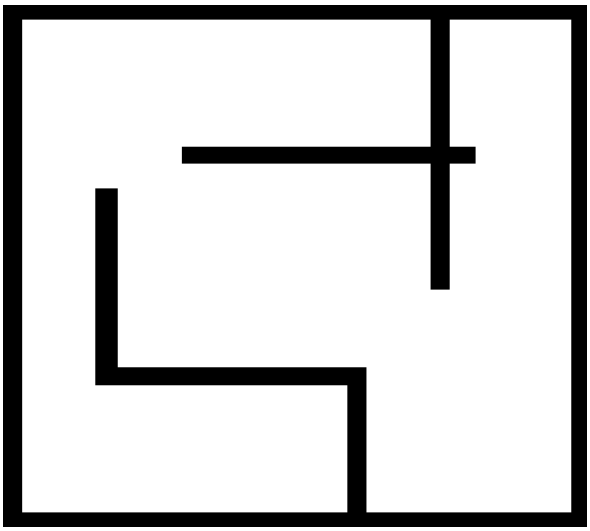


Fig. 4. The original map we used in the project.

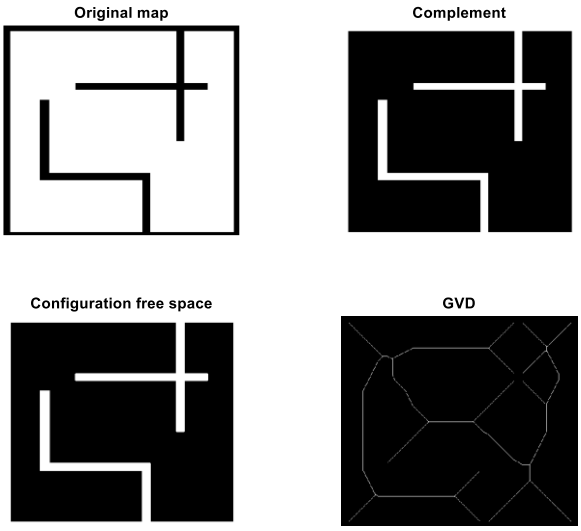


Fig. 5. The free space of the map in Voronoi diagram.

2) Multi-goal path planning

We chose local points P_n such as the points in Table.1 as a target (waypoints) similar to those in Fig. 3 by choosing its locations on the generated Voronoi diagram by adding its coordinates to the map. This stage we calculated the distant matrix using the equation

$$\sqrt{(p_{i+1,1} - p_{i,1})\text{^}2 + (p_{i+1,2} - p_{i,2})\text{^}2}$$

Then we combined all points in a map with the targets to find the optimal path to the local points using TSP. We formulated TSP algorithms by an integer linear programming and solved it by the Miller–Tucker–Zemlin (MTZ) formulation [2]. For time optimization, we used Goruobi modules as a solver for TSP.

3) Local navigator

DWA method will breakdown the optimal path, the green path specifically between waypoints, to multiple points that DWA can navigate locally.

Here is the last result where the robot will navigate to the local points and achieve the goal. We try to search the closest edge of the Voronoi diagram.

4) Mapping

III. CONCLUSION

References

APPENDIX. I